END SEM

(1) (b) Same as Mid-Sem (Q 2b)

(c) $f(z) = u + iv$

$u = ax^2 + bxy + cy^2$

According to CR condition, for $f(z)$ to be analytic;

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \& \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

$$\therefore \frac{\partial u}{\partial x} = 2ax + by = \frac{\partial v}{\partial y}$$

$$\Rightarrow \boxed{v(x,y) = \frac{by^2}{2} + 2axy + g(x) + c} \quad \text{— (i)}$$

$$\therefore \frac{\partial u}{\partial y} = bx + 2cy = -\frac{\partial v}{\partial x}$$

$$\Rightarrow \boxed{v(x,y) = -\frac{bx^2}{2} - 2cxy + h(y) + c} \quad \text{— (ii)}$$

Comparing (i) & (ii):

$$g(x) = -\frac{bx^2}{2} \; ; \quad h(y) = \frac{by^2}{2} \; ; \quad a = -c$$

$$\boxed{v(x,y) = -\frac{bx^2}{2} + 2axy - \frac{bx^2}{2} + c}$$

Hence, condition is $a = -c \in \mathbb{R}$; $b \in \mathbb{R}$

(Ans)

(a) $f(z) = \frac{1}{z-3}$

@ $z = 3$; $f(0) = -\frac{1}{3}$ $\rightarrow$ No singularity.

10

$z = 0$ is an <u>ordinary point</u> (Ans)

@ $z \rightarrow \infty$

Let $g(w) = f\left(\frac{1}{w}\right) = \frac{1}{\frac{1}{w} - 3} = \frac{w}{1-3w}$

$g(w)$ @ $w = 0 \Rightarrow g(0) = 0$

Hence, $g(w)$ is analytic near $w = 0$ $\rightarrow w = 0$

is a removable singularity for $g(w)$

Hence, $z = \infty$ is a <u>removable singularity</u> for $f(z)$ (Ans)

② (a) Same as MIDSEM (Q4 b)

(c) $z = e^{i\theta} = \cos\theta + i\sin\theta$.

$\therefore e^{3i\theta} = \cos 3\theta + i\sin 3\theta$

$(z + z^{-1})^3 = z^3 + z^{-3} + 3zz^{-1}(z + z^{-1})$

$\Rightarrow (e^{i\theta} + e^{-i\theta})^3 = e^{3i\theta} + e^{-3i\theta} + 3(e^{i\theta} + e^{-i\theta})$

$\Rightarrow (2\cos\theta)^3 = 2\cos 3\theta + 3 \times 2\cos\theta$

$\Rightarrow \boxed{4\cos^3\theta = \cos 3\theta + 3\cos\theta}$ (Proved)

(b) ③ $f(x)$ is piecewise continuous in $[-\pi, \pi]$.

" " single-valued in "

" " finite " "

$\therefore f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx$

$\therefore \int_c^{c+2\pi} f(x) \cos mx \, dx$

$= \frac{a_0}{2} \int_c^{c+2\pi} \cos mx + \sum_{n=1}^{\infty} a_n \int_c^{c+2\pi} \cos nx \cos mx \, dx$

$+ \sum_{n=1}^{\infty} b_n \int_c^{c+2\pi} \sin nx \cos mx \, dx$

Now, $\int_c^{c+2\pi} \cos mx \cos nx \, dx = \begin{cases} 0 ; & m \neq n \\ \pi ; & m = n \end{cases}$ (ii)

$\quad$ (i) $\quad = \int_c^{c+2\pi} \sin nx \sin mx \, dx$

$\int_c^{c+2\pi} \sin mx \cos nx \, dx = 0$ (ii) ; $\int_c^{c+2\pi} \cos nx \, dx$ — (iii)

$= \int_c^{c+2\pi} \sin nx \, dx = 0$

$\int_c^{c+2\pi} f(x) \cos mx \, dx = \sum_{n=1}^{\infty} a_n \int_c^{c+2\pi} \cos nx \cos mx \, dx$ [using (ii) & (iii)].

$= a_m \pi$ [using (i)]

$$\Rightarrow \boxed{a_m = \frac{1}{\pi} \int_c^{c+2\pi} f(x) \cos mx \, dx} \quad \text{— (iv)}.$$

Similarly,

$$\boxed{b_m = \frac{1}{\pi} \int_c^{c+2\pi} f(x) \sin mx \, dx} \quad \text{— (v)}$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx$$

$$\Rightarrow \int_c^{c+2\pi} f(x) \, dx = \int_c^{c+2\pi} \frac{a_0}{2} \, dx + \sum a_n \cancel{\int_c^{c+2\pi} \cos nx \, dx}^{\to 0}$$

$$+ \sum b_n \cancel{\int_c^{c+2\pi} \sin nx \, dx}^{\to 0}$$

[using (iii)]

$$= \frac{a_0}{2} \cdot 2\pi = a_0 \pi$$

$$\Rightarrow \boxed{a_0 = \frac{1}{\pi} \int_c^{c+2\pi} f(x) \, dx} \quad \text{— (vi)}$$

In our question :

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \, dx$$

$$= \frac{1}{\pi} \left[\int_{-\pi}^{0} \left(1 + \frac{2x}{\pi}\right) dx + \int_0^{\pi} \left(1 - \frac{2x}{\pi}\right) dx \right]$$

$$= \frac{1}{\pi} \left[x + \frac{x^2}{\pi} \Big|_{-\pi}^{0} + x - \frac{x^2}{\pi} \Big|_0^{\pi} \right]$$

$$= \frac{1}{\pi} \left[\pi - \pi + \pi - \pi \right] = 0$$

$$\boxed{a_0 = 0} \quad \text{— (vii)}$$

$$\therefore a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx \quad [\text{from (IV)}]$$

$$= \frac{1}{\pi}\left[\int_{-\pi}^{0}\left(1+\frac{2x}{\pi}\right)\cos nx\,dx + \int_{0}^{\pi}\left(1-\frac{2x}{\pi}\right)\cos nx\,dx\right]$$

$$= \frac{1}{\pi}\left[\frac{\sin nx}{n}\Big|_{-\pi}^{0} + \frac{2}{\pi}\int_{-\pi}^{0} x\cos nx\,dx + \frac{\sin nx}{n}\Big|_{0}^{\pi} - \frac{2}{\pi}\int_{0}^{\pi} x\cos nx\,dx\right].$$

$$= \frac{1}{\pi}\cdot\frac{2}{\pi}\left[\cancel{x\frac{\sin nx}{n}\Big|_{-\pi}^{0}}^{\,0} - \int_{-\pi}^{0}\frac{\sin nx}{n}\,dx - \cancel{x\frac{\sin nx}{n}\Big|_{0}^{\pi}}^{\,0} + \int_{0}^{\pi}\frac{\sin nx}{n}\,dx\right]$$

$$= \frac{2}{\pi^2}\left[-\frac{\cos nx}{n^2}\Big|_{0}^{\pi} + \frac{\cos nx}{n^2}\Big|_{-\pi}^{0}\right]$$

$$= \frac{2}{n^2\pi^2}\left[-\cos n\pi + 1 + 1 - \cos n\pi\right]$$

$$= \frac{4}{n^2\pi^2}(1-\cos n\pi)$$

$\underline{n = \text{even}} \longrightarrow \cos n\pi = 1 \longrightarrow a_n = 0$

$\underline{n = \text{odd}} \longrightarrow \cos n\pi = -1 \longrightarrow a_n = \frac{8}{n^2\pi^2}$

$$\boxed{a_n = \begin{cases} 0 & ; \ n = \text{even} \\ \dfrac{8}{n^2\pi^2} & ; \ n = \text{odd} \end{cases}} \quad \text{— (VIII)}$$

$$\therefore b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\,dx \quad [\text{from (V)}]$$

$$= \frac{1}{\pi}\left[\int_{-\pi}^{0}\left(1+\frac{2x}{\pi}\right)\sin nx\,dx + \int_{0}^{\pi}\left(1-\frac{2x}{\pi}\right)\sin nx\,dx\right]$$

$$= \frac{1}{\pi}\left[-\frac{\cos nx}{n}\Big|_{-\pi}^{0} + \frac{2}{\pi}\int_{-\pi}^{0} x\sin nx\,dx - \frac{\cos nx}{n}\Big|_{0}^{\pi} - \frac{2}{\pi}\int_{0}^{\pi} x\sin nx\,dx\right]$$

$$= \frac{1}{\pi}\left[\frac{1}{n}\left(\cos n\pi - 1 - \cos n\pi + 1\right) + \frac{2}{\pi}\left(x \frac{\cos nx}{n}\Big|_{-\pi}^{0} + \int_{-\pi}^{0} \frac{\cos nx}{n}\,dx - x\frac{\cos nx}{n}\Big|_{0}^{\pi} - \int_{0}^{\pi} \frac{\cos nx}{n}\,dx\right)\right]$$

$$= \frac{2}{\pi^2}\left(+\pi\frac{\cos n\pi}{n} + \frac{\sin nx}{n}\Big|_{-\pi}^{0} - \pi\frac{\cos n\pi}{n} - \frac{\sin nx}{n}\Big|_{0}^{\pi}\right)$$

$$\boxed{b_n = 0} \quad \text{— (IX)}$$

$$\therefore f(x) = \sum_{n=0}^{\infty} a_{2n+1} \cos\left[(2n+1)x\right]$$

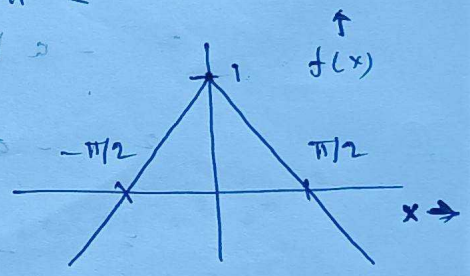
$$\cancel{\boxed{f(x) = \sum_{n=0}^{\infty} \frac{8}{n^2\pi^2}\cos nx}} \quad \text{(Ans)} \Rightarrow \boxed{f(x) = \sum_{n=0}^{\infty} \frac{8}{\left[(2n+1)\pi\right]^2}\cos\left[(2n+1)x\right]}$$

$$= \frac{8}{\pi^2}\left[\frac{\cos x}{1^2} + \frac{\cos 3x}{3^2} + \dots\right] \quad \text{(Ans)}$$

ALTERNATIVE :

$$f(x) = \begin{cases} 1 - \frac{2x}{\pi} ; & 0 \le x \le \pi \\ 1 + \frac{2x}{\pi} ; & -\pi \le x < 0 \end{cases}$$

Plotting $f(x)$ it can be seen that $f(x)$ is an even f^n.



$$\therefore f(x) = f(-x)$$

$$\therefore b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\,dx = \boxed{0 = b_n}$$

[$f(x) \to$ even ; $\sin nx \to$ odd ; $f(x)\sin nx \to$ odd]

$$\therefore a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\,dx$$

$$= \frac{1}{\pi}\int_{0}^{\pi} f(x)\,dx = \frac{2}{\pi}\int_{0}^{\pi} 1 - \frac{2x}{\pi}\,dx$$

$$= \frac{2}{\pi}\left[x - \frac{x^2}{\pi}\right]_0^{\pi}$$

$$\boxed{a_0 = 0}$$

$$\therefore a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx = \frac{2}{\pi}\int_{0}^{\pi} f(x)\cos nx\,dx$$

$$= \frac{2}{\pi} \int \left(1 - \frac{2x}{\pi}\right) \cos nx \, dx$$

$$= \frac{2}{\pi} \left[- \frac{\sin nx}{n} \Big|_0^{\pi} - \frac{2}{\pi} \int_0^{\pi} x \cos nx \, dx \right]$$

$$= 0 - \frac{4}{\pi^2} \left[- x \frac{\sin nx}{n} \Big|_0^{\pi} + \int \frac{\sin nx}{n} \, dx \right]$$

$$= - \frac{4}{\pi^2} \frac{\cos nx}{n^2} \Big|_0^{\pi} = - \frac{4}{n^2 \pi^2} (\cos n\pi - 1)$$

$$\therefore \boxed{a_n = \begin{cases} 0 \; ; & n = \text{even} \\ \frac{8}{n^2 \pi^2} \; ; & n = \text{odd} \end{cases}}$$ (Ans)

(4) $$F(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) \, e^{+ikx} \, dx$$

$$= \frac{1}{\sqrt{2\pi}} \int x \, e^{-ax^2} \cdot e^{ikx} \, dx$$

$$= \frac{1}{\sqrt{2\pi}} \int x \, e^{-ax^2 + ikx} \, dx$$

Now, $-ax^2 + ikx$

$$= -a \left[x^2 - \frac{ikx}{a} \right]$$

$$= -a \left[x^2 - \frac{2ikx}{2a} + \left(\frac{ik}{2a}\right)^2 - \left(\frac{ik}{2a}\right)^2 \right]$$

$$= -a \left[\left(x - \frac{ik}{2a}\right)^2 + \frac{k^2}{4a^2} \right]$$

$$= -a \left(x - \frac{ik}{2a}\right)^2 - \frac{k^2}{4a}$$

$$\therefore F(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} x \exp\left[-a \left(x - \frac{ik}{2a}\right)^2 - \frac{k^2}{4a} \right] dx$$

$$= \frac{e^{-k^2/4a}}{\sqrt{2\pi}} \int x \, e^{-a\left(x - \frac{ik}{2a}\right)^2} \, dx$$

Let $x - \frac{ik}{2a} = p$

$\Rightarrow dx = dp$

$$\therefore F(k) = \boxed{\frac{e^{-k^2/4a}}{\sqrt{2\pi}}} \int_{-\infty}^{\infty} \left(p + \frac{ik}{2a}\right) e^{-ap^2} \, dp$$

$$= \alpha \left[\underbrace{\int_{-\infty}^{\infty} p \, e^{-ap^2} \, dp}_{\text{(i)}} + \frac{ik}{2a} \underbrace{\int e^{-ap^2} \, dp}_{\text{(ii)}} \right]$$

$\to 0$ (odd func)

Now, $\int_{-\infty}^{\infty} e^{-ax^2}\, dx = \sqrt{\frac{\pi}{a}}$ ~~$= dI(a)$~~

~~$\Rightarrow \frac{dI(a)}{da} =$~~ For ① ; $u = p^2$

$\Rightarrow du = 2p\, dp.$

~~$\therefore \alpha \left[\frac{i}{2} \int_{-\infty}^{\infty} du\, e^{-au} + \frac{ik}{2a} \sqrt{\frac{\pi}{a}} \right].$~~

$F(k) = \alpha \left[\frac{ik}{2a} \cdot \sqrt{\frac{\pi}{a}} \right].$ (15)

$= \frac{e^{-k^2/4a}}{\sqrt{2\pi}} \cdot \frac{ik}{2a} \cdot \sqrt{\frac{\pi}{a}}$

$$\boxed{F(k) = \frac{ik\, e^{-k^2/4a}}{2\,(2a)^{3/2}}}$$ (Ans)

(5) $f(x, y) = (x^2 - y^2) \exp[-(x^2 + y^2)]$

$x = r\cos\theta$; $y = r\sin\theta$

$\therefore x^2 - y^2 = r^2(\cos^2\theta - \sin^2\theta) = r^2 \cos 2\theta$

$\therefore \boxed{f(r, \theta) = r^2 \cos 2\theta \exp(-r^2)}$ — ①

$\therefore$ ~~∇f~~ For polar coordinates :

$$\boxed{|\nabla f|^2 = \left(\frac{\partial f}{\partial r}\right)^2 + \frac{1}{r^2}\left(\frac{\partial f}{\partial \theta}\right)^2}$$

$\therefore \frac{\partial f}{\partial r} = 2r\cos 2\theta\, e^{-r^2} + 2\cos 2\theta\, r^3 e^{-r^2}$

$= 2r\cos 2\theta\, e^{-r^2}(1 - r^2)$

On the circle, $r = 1$.

$\therefore \boxed{\left.\frac{\partial f}{\partial r}\right|_{r=1} = 0}$

$\therefore \frac{\partial f}{\partial \theta} = -r^2 e^{-r^2}\, 2\sin 2\theta$

$\therefore \boxed{\left.\frac{\partial f}{\partial \theta}\right|_{r=1} = -2e^{-1}\sin 2\theta}$

$\therefore |\nabla f|^2_{r=1} = \frac{(-2e^{-1}\sin 2\theta)^2}{1^2} = \frac{4e^{-2}\sin^2 2\theta}{1^2}$

$\Rightarrow \boxed{|\nabla f|_{r=1} = 2e^{-1}\,|\sin 2\theta|}$

$\therefore \boxed{|V|_{max} = \frac{2}{e}}$ (Ans)

The maximum occurs when $|\sin 2\theta| = 1$

$\Rightarrow \sin 2\theta = \pm 1$

$\Rightarrow 2\theta = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \ldots$

$\Rightarrow \boxed{\theta = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \ldots}$

$\therefore x = r\cos\theta = \cos\theta$

$\therefore x_1 = \cos\frac{\pi}{4} = \frac{1}{\sqrt{2}}$

$x_2 = \cos\frac{3\pi}{4} = -\frac{1}{\sqrt{2}}$

$\therefore y = r\sin\theta = \sin\theta$

$\therefore y_1 = \sin\frac{\pi}{4} = \frac{1}{\sqrt{2}}$

$\therefore y_2 = \sin\frac{3\pi}{4} = \frac{1}{\sqrt{2}} = y_1.$

$\therefore y_3 = \sin\frac{5\pi}{4} = -\frac{1}{\sqrt{2}}$

Locat^ns on the circle are : $\left(\pm\frac{1}{\sqrt{2}}, \pm\frac{1}{\sqrt{2}}\right)$ (Ans)

$\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right), \left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right), \left(-\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right), \left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$

(6) $\frac{da}{dt} = 4\dot{a} + 3a$; $\frac{dx}{dt} = v$; $\frac{dv}{dt} = a$

Combining these equat^n

$$\frac{d}{dt}\underbrace{\begin{bmatrix} x \\ v \\ a \end{bmatrix}}_{X} = \underbrace{\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 34 & 43 \end{bmatrix}}_{M} \underbrace{\begin{bmatrix} x \\ v \\ a \end{bmatrix}}_{X}$$

$\Rightarrow \frac{dX}{dt} = MX$

$\Rightarrow \int_{X(0)}^{X(t)} \frac{dx}{x} = M\int_0^t dt$

$\Rightarrow \ln\frac{X(t)}{X(0)} = Mt$

$\Rightarrow \boxed{X(t) = e^{Mt} X(0)}$

$$X(0) = \begin{bmatrix} x(0) \\ v(0) \\ a(0) \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \\ 16 \end{bmatrix}$$

$$\therefore \quad M X(0) = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 4 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 4 \\ 16 \end{bmatrix}$$

$$= \begin{bmatrix} 4 \\ 16 \\ 64 \end{bmatrix} = 4 \begin{bmatrix} 1 \\ 4 \\ 16 \end{bmatrix} = 4 X(0)$$

$$\therefore \quad \boxed{M X(0) = 4 X(0)}$$

4 is an eigen~~vector~~ value of M, with eigenvector $X(0)$

$$\therefore \quad e^{Mt} X(0) = e^{4t} X(0) = X(t)$$

$$\Rightarrow \quad X(t) = e^{4t} \begin{bmatrix} 1 \\ 4 \\ 16 \end{bmatrix} = \begin{bmatrix} x(t) \\ v(t) \\ a(t) \end{bmatrix}$$

$$\Rightarrow \quad \boxed{\begin{aligned} x(t) &= e^{4t} \\ v(t) &= 4e^{4t} \\ a(t) &= 16e^{4t} \end{aligned}}$$

(Ans)

$$\therefore \quad \frac{x(t)}{x(0)} = \frac{e^{4t}}{4} \; ; \quad \frac{v(t)}{v(0)} = \frac{4e^{4t}}{4} = e^{4t}$$

$$\therefore \quad \frac{a(t)}{a(0)} = \frac{16e^{4t}}{16} = e^{4t}$$

$$\therefore \quad \boxed{\frac{x(t)}{x(0)} = \frac{v(t)}{v(0)} = \frac{a(t)}{a(0)}}$$

(Proved)